

Multiple Choice (10 marks)

1. Describe the function of the lateral-line system in fishes.
 - (a) The lateral-line system in fishes helps with the digestion of food.
 - (b) The lateral-line system in fishes is used for excreting waste materials.
 - (c) The lateral-line system in fishes is involved in the secretion of hormones.
 - (d) The lateral-line system in fishes is used for breathing.
 - (e) The lateral-line system in fishes is used for vision.
2. Which of the following statements about trypsin is NOT true?
 - (a) It is an organic compound made of proteins.
 - (b) It is a catalyst that alters the rate of a reaction.
 - (c) It is operative over a wide pH range.
 - (d) The rate of catalysis is affected by the concentration of substrate.
3. Which of the following adaptive features would least likely be found in an animal living in a hot arid environment?
 - (a) High metabolic rate to generate heat
 - (b) Storage of water in fatty tissues
 - (c) Long loops of Henle to maximize water reabsorption
 - (d) Short loops of Henle to maximize water secretion
 - (e) Large amounts of body hair to trap heat
4. Which of the following statements about lipids is correct?
 - (a) Lipids are the primary source of energy for cellular processes.
 - (b) Lipids are soluble in water but insoluble in organic solvents.
 - (c) Phospholipids are water soluble.
 - (d) Steroids are lipids that consist of glycerol and fatty acids.
 - (e) Lipids make up most cell surface receptors.
5. Why does milk "when kept in a refrigerator?
 - (a) Milk spoils due to high temperatures
 - (b) Milk spoils exclusively due to the separation of fat content at lower temperatures
 - (c) Milk spoils because of thermophilic bacteria
 - (d) Psychrophilic bacteria cause milk to spoil in cool temperatures.
 - (e) Milk spoils as a result of exposure to light inside the refrigerator

True / False (10 marks)

Answer True or False

6. True or False: Pheromones are chemical substances released by organisms that trigger physiological or behavioral responses in conspecifics.
7. True or False: Polytene chromosomes are produced only during meiosis and are not visible under a light microscope.
8. True or False: Warning coloration is a form of camouflage used by prey to make themselves less visible to predators in their habitat.
9. True or False: Ultrasound examination is an effective method for determining a dog's hearing sensitivity to various sound frequencies.
10. True or False: It is estimated that around 90% of the human genome is involved in protein coding.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the mechanisms by which a newborn can acquire syphilis, focusing on the potential pathways of transmission from mother to child during pregnancy or childbirth.
12. Discuss the nature of behavior, examining whether behavior patterns are primarily inherited or subject to modification, and analyze the implications of your findings.

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